

Camels and relatives

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Artiodactyla
FAMILY	Camelidae
SPECIES	7

Camelids—camels and their relatives— have long, slender legs and a distinctive gait known as pacing, whereby the front and back legs on the same side move forward together in a rocking motion. Of the Old World camelids—the camels—only one species (the Bactrian camel, from the

border area between western China and Mongolia) now survives in the wild. A camel is able to drink up to a quarter of its weight at a time, and can store the water for several days. The New World members of this family, the guanaco and vicuña, are found wild in South America; their domestic descendants, the llama and alpaca, have been bred in the Andes since the time of the Inca civilization. All domesticated camelids are vital to human survival: they provide people with hair, milk, and transportation.

Anatomy

Camelids have a relatively small head, a long, thin neck, and a split upper lip. Camels have either one hump (the dromedary) or 2 humps (the Bactrian camel): these store fat that can be drawn on during lean times. All camelids have a thick coat that provides insulation against daytime heat and warmth during the cooler nights or at altitude. Unlike other hoofed mammals, camelids rest their weight not on their hooves but on the

undersides of the 2 digits on each foot, which are cushioned by a fatty pad. This is an adaptation to walking on sandy soil. Camelids are unique among mammals in that their blood cells can expand significantly (as much as 240 percent) when the camel drinks, allowing it to rehydrate safely. The camel can drink a huge amount of water relative to other mammals without the cells rupturing.

Social interaction

In the wild, camelids form groups that consist of one dominant male and a harem of females. “Surplus” males form bachelor bands. While the social systems of the South American camelids have not been studied in great detail, more is known about the way camels interact with each other. Both species go through an elaborate and dramatic ritual when a dominant male is faced with a challenger. First, the harem leader grinds his teeth, rubs a gland on the back of his head against his hump (or front hump in the case of the Bactrian camel), smacks his tail loudly against his rump, and urinates on his back legs, rump, and tail. The 2 then pace side by side, display their tall, humped profiles, and extrude a red bladderlike sac (the dulaa) from the corner of the mouth.

Wild and domestic

The only living Old World wild camelid is the Bactrian camel, although its numbers are now

reduced to only 1,000–2,000 individuals. It is taller and slimmer than its domestic counterpart and has more compact, pointed humps. Domestic Bactrian camels are used for transport in cold regions from northern China to Turkey. The dromedary is extinct in the wild, although feral herds now live in central Australia. Domestic dromedaries are found in hot regions in North and northeastern Africa, the Middle East, and northern India through to Kazakhstan. As with other domestic animals, there are different breeds, one of which is kept for its speed and is used in camel racing.

The llama is a domestic animal bred from the wild guanaco and is the traditional pack animal in the Andes. The alpaca is also a domestic species, bred for its fine wool. In the past, the alpaca was thought to be a descendant of the guanaco, but current genetic evidence indicates that it is in fact a descendant of the vicuña, the fine-fleeced, wild camelid of the high-altitude Andes.



DESERT SPECIALISTS

Camels, such as these wild bactrians, are well adapted to life in hot climates. Their broad feet provide stability on soft sand, and they have long eyelashes and slitlike, closable nostrils that afford protection during dust storms.